

Lecture 8: Forecasting

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Forecasting

Introduction

Recap: Procedure of econometric analysis

- Econometric analysis proceeds in five steps:
 1. Definition
 2. Data
 3. Identification
 4. Estimation
 5. Inference and/or **Forecasting**

Outline

- Preliminary: Statistics
- One-Period-Ahead Forecasting
- Multi-Period-Ahead Forecasting
- Impulse Response Function
- Model Selection

Preliminary

Statistics

Five-Step Procedure of (Two-Sided) Hypothesis Testing

(Step 1): Determine the hypothesis: $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$.

(Step 2): Specify the level of significance α .

(Step 3): Derive the sampling distribution of $\hat{\theta}$ and construct the test statistic t under H_0 .

(Step 4-1): Compute the critical value $t_{\alpha/2}$: $P(|t| < t_{\alpha/2}) = 1 - \alpha$.

(Step 4-2): Construct the $(1 - \alpha)$ level confidence interval:

$CI_\alpha(\hat{\theta}) := (\hat{\theta} - t_{\alpha/2}se(\hat{\theta}), \hat{\theta} + t_{\alpha/2}se(\hat{\theta}))$, where $se(\hat{\theta})$ is the standard error.

(Step 5): Reject H_0 if and only if $\theta_0 \notin CI_\alpha$.

- $P(|t| < t_{\alpha/2}) = 1 - \alpha$
 $\iff P(\hat{\theta} - t_{\alpha/2}se(\hat{\theta}) < \theta_0 < \hat{\theta} + t_{\alpha/2}se(\hat{\theta})) = 1 - \alpha$
 $\iff P(\theta_0 \in CI_\alpha(\hat{\theta})) = 1 - \alpha$
i.e., θ_0 is included in CI_α with probability $1 - \alpha$
 $\hookrightarrow CI_\alpha(\hat{\theta})$ can be viewed as an “interval estimate” of θ_0 .

Forecasting

One-Period-Ahead Forecasting

One-Period-Ahead Forecasting: Definition

- **Setup:** We have observed a univariate time-series $\{y_s\}_{s=1}^t$.
At (the end of) time t , we want to forecast its value at time $t + 1$, i.e., y_{t+1} .
- I_t : the information set available at (the end of) time t .
 - e.g., $I_t := \{y_0, y_1, \dots, y_t\}$.
- $f_{t+1|t}$: any forecast of y_{t+1} made at (the end of) time t .
- y_{t+1} is not yet realized at time t .
 - $\hookrightarrow$ Q. How should we assess the performance of $f_{t+1|t}$?

One-Period-Ahead Forecasting: Definition

- Idea: Minimize the loss from the forecast error.
- The optimal (point) forecast of y_{t+1} at time t :

$$y_{t+1|t} \in \arg \min_{f_{t+1|t}} E[(y_{t+1} - f_{t+1|t})^2 | I_t],$$

yielding

$$y_{t+1|t} = E[y_{t+1} | I_t].$$

- Q. Can we identify the object of interest $y_{t+1|t}$?

One-Period-Ahead Forecasting: Identification

- We need a model to link y_{t+1} and I_t .
- A linear forecasting model:

$$y_{t+1} = \beta_0 + \beta_1 y_t + \varepsilon_{t+1},$$

with the standard set of large-sample-theory assumptions.

1. The parameters β_0 and β_1 are identified.
2. Take expectation: $y_{t+1|t} = E[y_{t+1}|I_t] = \beta_0 + \beta_1 y_t + \underbrace{E[\varepsilon_{t+1}|I_t]}_0 = \beta_0 + \beta_1 y_t.$
 $\rightarrow y_{t+1|t}$ is identified.

One-Period-Ahead Forecasting: Estimation

- We consider estimating (i.e., forecasting) $y_{t+1|t}$ in two ways:
 - Point forecast: $\hat{y}_{t+1|t}$
 - Interval forecast: $CI_{\alpha}(\hat{y}_{t+1|t})$

One-Period-Ahead Forecasting: Estimation

- **Point forecast $\hat{y}_{t+1|t}$:**

(Step 1) Estimate the linear forecasting model:

- e.g., obtain $\hat{\beta}_0$ and $\hat{\beta}_1$

(Step 2) Plug these estimates back into the identification result:

- e.g., $\hat{y}_{t+1|t} = \hat{\beta}_0 + \hat{\beta}_1 y_t$

One-Period-Ahead Forecasting: Estimation

- **Interval forecast** $CI_\alpha(\hat{y}_{t+1|t})$:

$$P(y_{t+1} \in CI_\alpha(\hat{y}_{t+1|t})) = 1 - \alpha.$$

- Problem: The probability distribution reflects two sources of randomness:
 1. The sampling variation of $\hat{y}_{t+1|t}$
 2. As-yet-realized future fluctuation in y_{t+1}
- Idea: Look at the forecast error:

$$\hat{e}_{t+1|t} := y_{t+1} - \hat{y}_{t+1|t}.$$

One-Period-Ahead Forecasting: Estimation

- The forecast error:

$$\hat{e}_{t+1|t} := y_{t+1} - \hat{y}_{t+1|t} = (\beta_0 + \beta_1 y_t + \varepsilon_{t+1}) - (\hat{\beta}_0 + \hat{\beta}_1 y_t).$$

- Mean:

$$E[\hat{e}_{t+1|t}|I_t] = (\beta_0 + \beta_1 y_t + \underbrace{E[\varepsilon_{t+1}|I_t]}_0) - (\underbrace{E[\hat{\beta}_0|I_t]}_{\xrightarrow{p} \beta_0} + \underbrace{E[\hat{\beta}_1|I_t]}_{\xrightarrow{p} \beta_1} y_t) = 0.$$

- Variance:

$$\text{Var}(\hat{e}_{t+1|t}|I_t) = \underbrace{\text{Var}(y_{t+1}|I_t)}_{\sigma_\varepsilon^2} + \text{Var}(\hat{y}_{t+1|t}|I_t) + \underbrace{\text{Cov}(y_{t+1}, \hat{y}_{t+1|t}|I_t)}_0 = \sigma_\varepsilon^2 + \text{Var}(\hat{y}_{t+1|t}|I_t).$$

In practice,

$$\widehat{\text{Var}}(\hat{e}_{t+1|t}|I_t) := \hat{\sigma}_\varepsilon^2 + \widehat{\text{Var}}(\hat{y}_{t+1|t}|I_t).$$

One-Period-Ahead Forecasting: Estimation

- Further assume

$$\varepsilon_s \sim GWN(0, \sigma_\varepsilon^2) \quad \forall s = 1, \dots, t, t+1.$$

- Then,

$$\hat{e}_{t+1|t} := y_{t+1} - \hat{y}_{t+1|t} = (\beta_0 + \beta_1 y_t + \varepsilon_{t+1}) - (\hat{\beta}_0 + \hat{\beta}_1 y_t) \sim \mathcal{N}(0, \text{Var}(\hat{e}_{t+1|t}|I_t))$$

so that

$$\frac{\hat{e}_{t+1|t}}{(\text{Var}(\hat{e}_{t+1|t}|I_t))^{1/2}} \sim \mathcal{N}(0, 1)$$

and

$$\frac{\hat{e}_{t+1|t}}{\text{se}(\hat{e}_{t+1|t})} \sim t(T - K),$$

where $\text{se}(\hat{e}_{t+1|t}) := (\widehat{\text{Var}}(\hat{e}_{t+1|t}|I_t))^{1/2}$.

One-Period-Ahead Forecasting: Estimation

- Hence,

$$P\left(\left|\frac{\hat{e}_{t+1|t}}{se(\hat{e}_{t+1|t})}\right| < t_{\alpha/2}\right) = 1 - \alpha$$

$$\iff P(\hat{y}_{t+1|t} - t_{\alpha/2}se(\hat{e}_{t+1|t}) < y_{t+1} < \hat{y}_{t+1|t} + t_{\alpha/2}se(\hat{e}_{t+1|t})) = 1 - \alpha$$

$$\iff P(y_{t+1} \in CI_{\alpha}(\hat{y}_{t+1|t})) = 1 - \alpha,$$

where $CI_{\alpha}(\hat{y}_{t+1|t}) := (\hat{y}_{t+1|t} - t_{\alpha/2}se(\hat{e}_{t+1|t}), \hat{y}_{t+1|t} + t_{\alpha/2}se(\hat{e}_{t+1|t}))$.

Forecasting

Multiple-Period-Ahead Forecasting

Multiple-Period-Ahead Forecasting: Definition

- $f_{t+h|t}$: any forecast of y_{t+h} made at (the end of) time t
- The optimal (point) forecast of y_{t+h} at time t :

$$y_{t+h|t} \in \arg \min_{f_{t+h|t}} E[(y_{t+h} - f_{t+h|t})^2 | I_t],$$

yielding

$$y_{t+h|t} = E[y_{t+h} | I_t].$$

- Q. Can we identify the object of interest $y_{t+h|t}$?

Multiple-Period-Ahead Forecasting: Identification

- We need a model to link y_{t+h} and I_t .
- A linear forecasting model:

$$y_{t+1} = \beta_0 + \beta_1 y_t + \varepsilon_{t+1},$$

with the standard set of large-sample-theory assumptions.

1. The parameters β_0 and β_1 are identified.
2. By repeated substitution, $y_{t+h} = \beta_0 \sum_{j=0}^{h-1} \beta_1^j + \beta_1^h y_t + \sum_{j=0}^{h-1} \beta_1^j \varepsilon_{t+h-j}$.
3. Take expectation:

$$y_{t+h|t} = E[y_{t+h}|I_t] = \beta_0 \sum_{j=0}^{h-1} \beta_1^j + \beta_1^h y_t + \sum_{j=0}^{h-1} \beta_1^j \underbrace{E[\varepsilon_{t+h-j}|I_t]}_0 = \beta_0 \sum_{j=0}^{h-1} \beta_1^j + \beta_1^h y_t.$$

$\rightarrow y_{t+h|t}$ is identified.

Multiple-Period-Ahead Forecasting: Estimation

- We consider estimating (i.e., forecasting) $y_{t+h|t}$ in two ways:
 - Point forecast: $\hat{y}_{t+h|t}$
 - Interval forecast: $CI_{\alpha}(\hat{y}_{t+h|t})$

Multiple-Period-Ahead Forecasting: Estimation

- **Point forecast $\hat{y}_{t+h|t}$:**

(Step 1) Estimate the time-series model:

- e.g., obtain $\hat{\beta}_0$ and $\hat{\beta}_1$

(Step 2) Plug these estimates back into the identification result:

- e.g., $\hat{y}_{t+h|t} = \hat{\beta}_0 \sum_{j=0}^{h-1} \hat{\beta}_1^j + \hat{\beta}_1^h y_t$

Multiple-Period-Ahead Forecasting: Estimation

- **Interval forecast** $CI_\alpha(\hat{y}_{t+h|t})$:

$$P(y_{t+h} \in CI_\alpha(\hat{y}_{t+h|t})) = 1 - \alpha.$$

- Problem: The probability distribution reflects two sources of randomness:
 1. The sampling variation of $\hat{y}_{t+h|t}$
 2. As-yet-realized future fluctuation in y_{t+h}
- Idea: Look at the forecast error:

$$\hat{e}_{t+h|t} := y_{t+h} - \hat{y}_{t+h|t}.$$

Multiple-Period-Ahead Forecasting: Estimation

- The forecast error:

$$\hat{e}_{t+h|t} := y_{t+h} - \hat{y}_{t+h|t} = \left(\beta_0 \sum_{j=0}^{h-1} \beta_1^j + \beta_1^h y_t + \sum_{j=0}^{h-1} \beta_1^j \varepsilon_{t+h-j} \right) - \left(\hat{\beta}_0 \sum_{j=0}^{h-1} \hat{\beta}_1^j + \hat{\beta}_1^h y_t \right).$$

- Mean:

$$E[\hat{e}_{t+h|t} | I_t] = \left(\beta_0 \sum_{j=0}^{h-1} \beta_1^j + \beta_1^h y_t + \sum_{j=0}^{h-1} \beta_1^j \underbrace{E[\varepsilon_{t+h-j} | I_t]}_0 \right) - \left(\underbrace{E\left[\hat{\beta}_0 \sum_{j=0}^{h-1} \hat{\beta}_1^j \middle| I_t \right]}_{\xrightarrow{p} \beta_0 \sum_{j=0}^{h-1} \beta_1^j} + \underbrace{E[\hat{\beta}_1^h | I_t]}_{\xrightarrow{p} \beta_1^h} y_t \right) = 0.$$

- Variance:

$$\text{Var}(\hat{e}_{t+h|t} | I_t) = \underbrace{\text{Var}(y_{t+h} | I_t)}_{\sigma_\varepsilon^2} + \text{Var}(\hat{y}_{t+h|t} | I_t) + \underbrace{\text{Cov}(y_{t+h}, \hat{y}_{t+h|t} | I_t)}_0 = \sigma_\varepsilon^2 + \text{Var}(\hat{y}_{t+h|t} | I_t).$$

In practice,

$$\widehat{\text{Var}}(\hat{e}_{t+h|t} | I_t) := \hat{\sigma}_\varepsilon^2 + \widehat{\text{Var}}(\hat{y}_{t+h|t} | I_t).$$

Multiple-Period-Ahead Forecasting: Estimation

- Further assume

$$\varepsilon_s \sim GWN(0, \sigma_\varepsilon^2) \quad \forall s = 1, \dots, t, t+1.$$

- Then,

$$\hat{e}_{t+h|t} := y_{t+h} - \hat{y}_{t+h|t} \sim \mathcal{N}(0, \text{Var}(\hat{e}_{t+h|t}|I_t))$$

so that

$$\frac{\hat{e}_{t+h|t}}{(\text{Var}(\hat{e}_{t+h|t}|I_t))^{1/2}} \sim \mathcal{N}(0, 1)$$

and

$$\frac{\hat{e}_{t+h|t}}{\text{se}(\hat{e}_{t+h|t})} \sim t(T-K),$$

where $\text{se}(\hat{e}_{t+h|t}) := (\widehat{\text{Var}}(\hat{e}_{t+h|t}|I_t))^{1/2}$.

Multiple-Period-Ahead Forecasting: Estimation

- Hence,

$$P\left(\left|\frac{\hat{e}_{t+h|t}}{se(\hat{e}_{t+h|t})}\right| < t_{\alpha/2}\right) = 1 - \alpha$$

$$\iff P(\hat{y}_{t+h|t} - t_{\alpha/2}se(\hat{e}_{t+h|t}) < y_{t+h} < \hat{y}_{t+h|t} + t_{\alpha/2}se(\hat{e}_{t+h|t})) = 1 - \alpha$$

$$\iff P(y_{t+h} \in CI_{\alpha}(\hat{y}_{t+h|t})) = 1 - \alpha,$$

where $CI_{\alpha}(\hat{y}_{t+h|t}) := (\hat{y}_{t+h|t} - t_{\alpha/2}se(\hat{e}_{t+h|t}), \hat{y}_{t+h|t} + t_{\alpha/2}se(\hat{e}_{t+h|t}))$.

Multiple-Period-Ahead Forecasting: Caveat

- These results are derived under the large-sample setup.
- In a finite (or small) sample, the OLS estimator $\hat{\beta}$ is biased:
 $\implies$ The point and interval forecasts are biased.
- The biases tend to be large as the forecast horizon h increases.
 - e.g., $E[\hat{\beta}|I_t] \neq \beta$ vs. $E[\hat{\beta}^h|I_t] \neq \beta^h$

Forecasting

Impulse Response Function

Impulse Response Function: Definition

- **Insight:** $I_t := \{y_0, y_1, \dots, y_{t-1}, y_t\}$ with an AR(1) model $y_t = \rho_0 + \rho_1 y_{t-1} + \varepsilon_t$
 $\iff I_t := \{y_0, y_1, \dots, y_{t-1}, \varepsilon_t\}$ with an AR(1) model $y_t = \rho_0 + \rho_1 y_{t-1} + \varepsilon_t$
 $\hookrightarrow y_{t+h|t} = E[y_{t+h}|I_t]$ depends on ε_t .
- “What would have happened if $\varepsilon_t + \alpha$ occurred instead of ε_t ?”
- Impulse response function:

$$\begin{aligned} IRF_t(h) &:= \frac{\partial E[y_{t+h}|I_t]}{\partial \varepsilon_t} \\ &= \lim_{\alpha \rightarrow 0} \frac{E[y_{t+h}|I_{t-1} \cup \{\varepsilon_t + \alpha\}] - E[y_{t+h}|I_{t-1} \cup \{\varepsilon_t\}]}{\alpha}. \end{aligned}$$

i.e., the responsiveness of $y_{t+h|t}$ with respect to a marginal change in ε_t .

Impulse Response Function: Definition

- Cumulative impulse response function:

$$CIRF_t(h) := \sum_{\ell=1}^h IRF_t(\ell)$$

i.e., the accumulated effect of the change in ε_t from time t to $t + h$.

- Long-run impulse response function:

$$LRIRF_t(h) := \lim_{h \rightarrow \infty} CIRF_t(h)$$

i.e., the full effect of the change in ε_t over all time.

Impulse Response Function: Examples

- Ex1) The one-step impulse response function:

$$IRF_t(1) := \frac{\partial E[y_{t+1}|I_t]}{\partial \varepsilon_t} = \lim_{\alpha \rightarrow 0} \frac{E[y_{t+1}|I_{t-1} \cup \{\varepsilon_t + \alpha\}] - E[y_{t+1}|I_{t-1} \cup \{\varepsilon_t\}]}{\alpha}.$$

An AR(1) model: $y_t = \rho_0 + \rho_1 y_{t-1} + \varepsilon_t$.

$$\implies y_{t+1} = \rho_0 + \rho_1 y_t + \varepsilon_{t+1}.$$

$$- E[y_{t+1}|I_{t-1} \cup \{\varepsilon_t\}] = \rho_0 + \rho_1 y_t = \rho_0(1 + \rho_1) + \rho_1^2 y_{t-1} + \rho_1 \varepsilon_t.$$

$$- E[y_{t+1}|I_{t-1} \cup \{\varepsilon_t + \alpha\}] = \rho_0 + \rho_1 y_t = \rho_0(1 + \rho_1) + \rho_1^2 y_{t-1} + \rho_1(\varepsilon_t + \alpha).$$

Hence,

$$IRF_t(1) = \lim_{\alpha \rightarrow 0} \frac{\rho_1 \alpha}{\alpha} = \rho_1.$$

$\hookrightarrow IRF_t(1)$ is identified.

$\hookrightarrow IRF_t(1)$ can be estimated by $\widehat{IRF}_t(1) = \widehat{\rho}_1$.

Impulse Response Function: Examples

- Ex2) The h-step impulse response function:

$$IRF_t(h) := \frac{\partial E[y_{t+h}|I_t]}{\partial \varepsilon_t} = \lim_{\alpha \rightarrow 0} \frac{E[y_{t+h}|I_{t-1} \cup \{\varepsilon_t + \alpha\}] - E[y_{t+h}|I_{t-1} \cup \{\varepsilon_t\}]}{\alpha}.$$

An AR(1) model: $y_t = \rho_0 + \rho_1 y_{t-1} + \varepsilon_t$.

$$\Rightarrow y_{t+h} = \rho_0 \sum_{j=0}^h \rho_1^j + \rho_1^{h+1} y_{t-1} + \rho_1^h \varepsilon_t + \sum_{j=0}^{h-1} \rho_1^j \varepsilon_{t+h-j}.$$

$$- E[y_{t+h}|I_{t-1} \cup \{\varepsilon_t\}] = \rho_0 \sum_{j=0}^h \rho_1^j + \rho_1^{h+1} y_{t-1} + \rho_1^h \varepsilon_t.$$

$$- E[y_{t+h}|I_{t-1} \cup \{\varepsilon_t + \alpha\}] = \rho_0 \sum_{j=0}^h \rho_1^j + \rho_1^{h+1} y_{t-1} + \rho_1^h (\varepsilon_t + \alpha).$$

Hence,

$$IRF_t(h) = \lim_{\alpha \rightarrow 0} \frac{\rho_1^h \alpha}{\alpha} = \rho_1^h.$$

$\hookrightarrow IRF_t(h)$ is identified.

$\hookrightarrow IRF_t(h)$ can be estimated by $\widehat{IRF}_t(h) = \widehat{\rho}_1^h$.

Impulse Response Function: Examples

- Ex3) The cumulative impulse response function:

$$CIRF_t(h) := \sum_{\ell=1}^h IRF_t(\ell) = \sum_{\ell=1}^h \rho_1^\ell = \frac{1 - \rho_1^h}{1 - \rho_1} - 1$$

$\hookrightarrow CIRF_t(h)$ is identified.

$\hookrightarrow CIRF_t(h)$ can be estimated by $\widehat{CIRF}_t(h) = \frac{1 - \widehat{\rho}_1^h}{1 - \widehat{\rho}_1} - 1$.

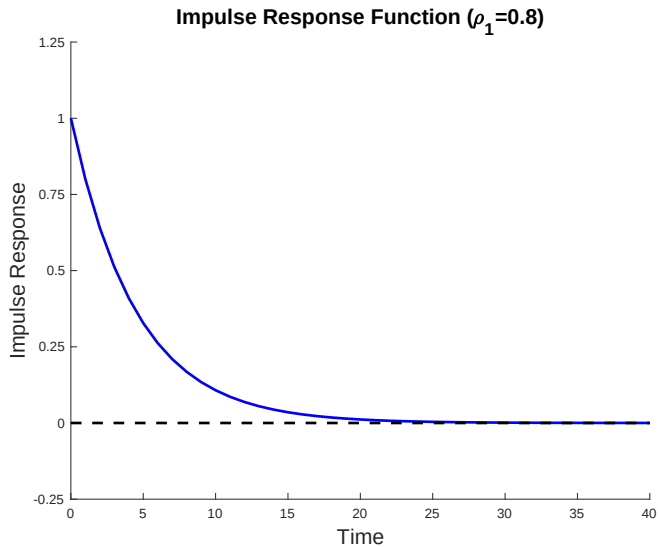
- Ex4) The Long-run impulse response function:

$$LRIRF_t(h) := \lim_{h \rightarrow \infty} CIRF_t(h) = \lim_{h \rightarrow \infty} \sum_{\ell=1}^h \rho_1^\ell = \frac{1}{1 - \rho_1} - 1 = \frac{\rho_1}{1 - \rho_1}$$

$\hookrightarrow LRIRF_t(h)$ is identified.

$\hookrightarrow LRIRF_t(h)$ can be estimated by $\widehat{LRIRF}_t(h) = \frac{\widehat{\rho}_1}{1 - \widehat{\rho}_1}$.

Impulse Response Function: Examples



Forecasting

Model Selection

Model Selection: Big Picture

- Three approaches for econometric modeling:
 1. Theory-driven approach
 - The model is guided by an economic theory.
 - Econometrics provides sophisticated ways to “bring data into line” with a particular theory.
 - “Structural approach”
 2. Data-driven approach
 - The model is designed to capture the characteristics of data.
 - No economic theory is used.
 - “Reduced-form approach”
 3. Hybrid approach
 - The model is motivated by an economic theory.
 - The model reflects key features of data.

Model Selection: Big Picture

- Modeling questions in practice:
 - Should my model include all the variables in the dataset?
 - How many explanatory variables do I need in my model?
 - How many lagged variables should I include?
 - Are the error terms really homoskedastic?
 - Should my model include a time trend or seasonality?
- etc.

Model Selection: Big Picture

- A good model should have the following features:
 1. Coherence with theory
 - The model has a theoretical underpinning.
 2. Coherence with data
 - The modeled $\hat{y}$ and the observed y have the same properties.
 - The model reflects all information about the data.
i.e., there is nothing left in the errors
 3. Validity of prediction/forecast
 - Prediction/forecast performs well.

Model Selection: Big Picture

- There are four (not mutually exclusive) categories:
 - (1) Test-based approach
 - (2) Goodness-of-fit approach
 - (3) Forecast-based approach
 - (4) Bayesian approach

Model Selection: Test-Based Approach

- The performance of a model is assessed by the appropriateness of the assumptions.
- We test econometric assumptions on the model.
 - e.g., tests for exogeneity, heteroskedasticity, autocorrelation, unit roots, etc.
- We test economic assumptions behind the model.
 - e.g., the propensity-to-consume is positive

Model Selection: Test-Based Approach

- Example:

(1) Begin with a model:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 z_t + \varepsilon_t$$

under TS1–TS5. Hope to learn about β_1 .

(2) Tests for TS1–TS5.

- e.g., exogeneity, heteroskedasticity, etc.

(3) Tests for an economic hypothesis.

- e.g., $H_0 : \beta_2 = 0$ vs. $H_1 : \beta_2 > 0$

(4) Modify the model appropriately.

- In modifying the model, we add and/or drop variables.
- The model has to be both parsimonious and consistent with the data.

Model Selection: Test-Based Approach

- There are two ways of doing this search.
- (i) Specific-to-general (or forward) method
 - The model starts simple.
 - The model expands as long as new variables do not break down the tests.
- (ii) General-to-specific (or backward) method
 - The model starts rich.
 - The model simplifies as the existing variables do not survive the tests.
 - e.g., the sequential t rule in the ADF test

Model Selection: Goodness-of-fit approach

- The performance of a model is assessed by some distance measures.
- The coefficient of determination:

$$R^2 := 1 - \frac{\sum_{t=1}^T (y_t - \hat{y}_t)^2}{\sum_{t=1}^T (y_t - \bar{y})^2}, \text{ where } \bar{y} := \frac{1}{T} \sum_{t=1}^T y_t.$$

i.e., the proportion of the dependent variable variance that is explained by the model.

- This measure always increases when a variable is added to the model.
 $\implies$ not a reliable measurement...

Model Selection: Goodness-of-fit approach

- Idea: Penalize the addition of variables by the degree of freedom.
- K : the number of regressors.
- The adjusted coefficient of determination:

$$\bar{R}^2 := 1 - \frac{\frac{1}{T-K} \sum_{t=1}^T (y_t - \hat{y}_t)^2}{\frac{1}{T-1} \sum_{t=1}^T (y_t - \bar{y})^2} = 1 - \frac{T-1}{T-K} (1 - R^2) \text{ where } \bar{y} := \frac{1}{T} \sum_{t=1}^T y_t,$$

with $T - K$ and $T - 1$ being the corresponding degrees of freedom.

Model Selection: Goodness-of-fit approach

- Idea: Measure the distance between the true density g and the estimated density f .
- Kullback-Leibler information criterion (KLIC):

$$KLIC(g, f) := \int g(y) \log\left(\frac{g(y)}{f(y)}\right) dy.$$

- When $f(y) = g(y)$, $KLIC(g, f)$ is minimized and equal 0.

Model Selection: Goodness-of-fit approach

- Akaike Information Criterion (AIC):

$$AIC(\hat{\beta}) := -2\log\mathcal{I}(\hat{\beta}) + 2K,$$

where K is the number of parameters.

- In finite sample, $\mathcal{I}(\hat{\beta}) := L_f(\hat{\beta})$, the likelihood function associated with f evaluated at $\hat{\beta}$.
- In large sample, $\mathcal{I}(\hat{\beta}) := MSE_f(\hat{\beta})$, the MSE (associated with f) evaluated at $\hat{\beta}$.
- AIC is a penalized log-likelihood (or log-MSE).
- The model with the lowest AIC has the highest estimated match with true density.
- There are variations of AIC:
 - e.g., BIC, DIC, TIC, etc.

Model Selection: Forecast-based approach

- The performance of a model is assessed by the performance of its forecasts.
- Two dimensions:
 1. Types of samples used for forecasting:
 - in-sample forecasting
 - out-of-sample forecasting
 2. Types of loss functions:
 - mean squared forecast errors (MSFE)
 - mean absolute forecast errors (MAFE)

Model Selection: Forecast-based approach

Types of samples used for forecasting

- In-sample forecasting:
 - (i) Use I_t (i.e., all information) to estimate $\hat{\rho}$.
 - (ii) Calculate the h -step-ahead forecast by

$$\hat{y}_{\ell+h|\ell} = \hat{\rho}_1 + \hat{\rho}_1 y_\ell$$

for all $\ell = 1, \dots, t - h$.

Model Selection: Forecast-based approach

Types of samples used for forecasting

- Out-of-sample forecasting:

(i) Divide I_t into two parts:

$$I_t = \underbrace{\{y_0, y_1, \dots, y_n\}}_{I_t^n}, \underbrace{\{y_{n+1}, \dots, y_t\}}_{I_t^m},$$

where $t = n + m$.

(ii) Use I_t^n (i.e., partial information) to estimate $\hat{\rho}$.

(iii) Calculate the h -step-ahead forecast by

$$\hat{y}_{n+h|n} = \hat{\rho}_1 + \hat{\rho}_1 y_n$$

(iv) Update $n := n + 1$ and repeat these until $n = t - h$.

Model Selection: Forecast-based approach

Types of loss functions

- mean squared forecast errors (MSFE):

$$MSFE_{t,n}(h) := \frac{1}{m-h} \sum_{\ell=n}^{t-h} (y_{\ell+h} - \hat{y}_{\ell+h|\ell})^2$$

↪ sensitive to outliers

- mean absolute forecast errors (MAFE):

$$MAFE_{t,n}(h) := \frac{1}{m-h} \sum_{\ell=n}^{t-h} |y_{\ell+h} - \hat{y}_{\ell+h|\ell}|$$

↪ robust to outliers

Summary and Preview

Summary

- We studied forecasting.
- Baseline theories of forecasting:
 - One-period-ahead forecasting
 - Multiple-period-ahead forecasting
- Two applications of forecasting:
 - Impulse response functions
 - Model selection

Preview

- We have focused on a univariate time-series model:
 - e.g., $y_t \leftarrow y_{t-1}$
- We will consider a multivariate time-series model:
 1. The contemporaneous relationship:
 - e.g., $y_t \leftrightarrow x_t$ (i.e., the bilateral feedback between y_t and x_t)
 2. The intertemporal relationship:
 - e.g., $y_t, x_t \leftarrow y_{t-1}, x_{t-1}$
- Vector Autoregressive (VAR) models
- **Next class:** April 23 (Thu), 11:00-12:15, Room 0.4, Blandijn, Campus Boekentoren
- **Reminder:** Assignment 2 is due by 11:00, April 23 (Thu).